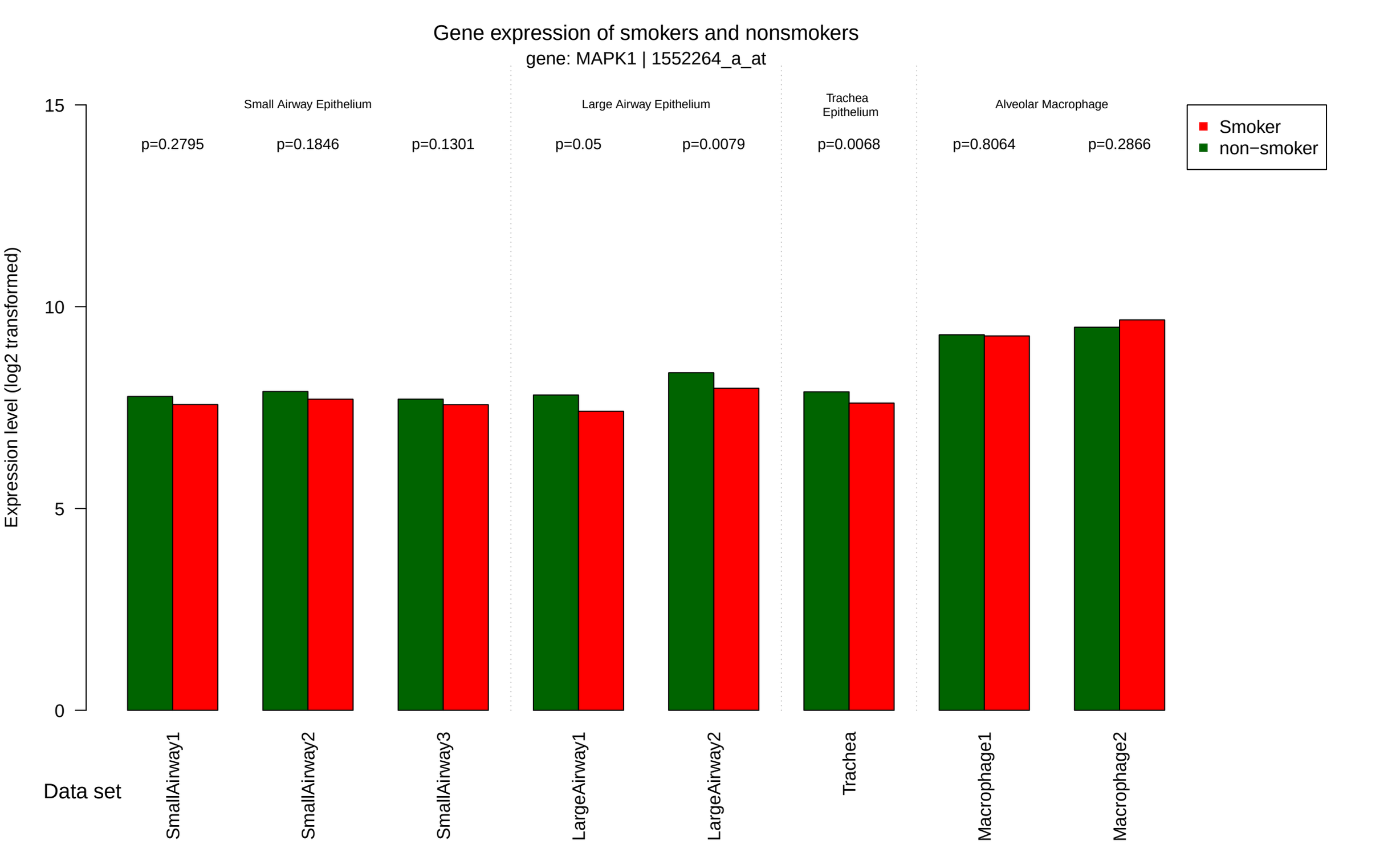
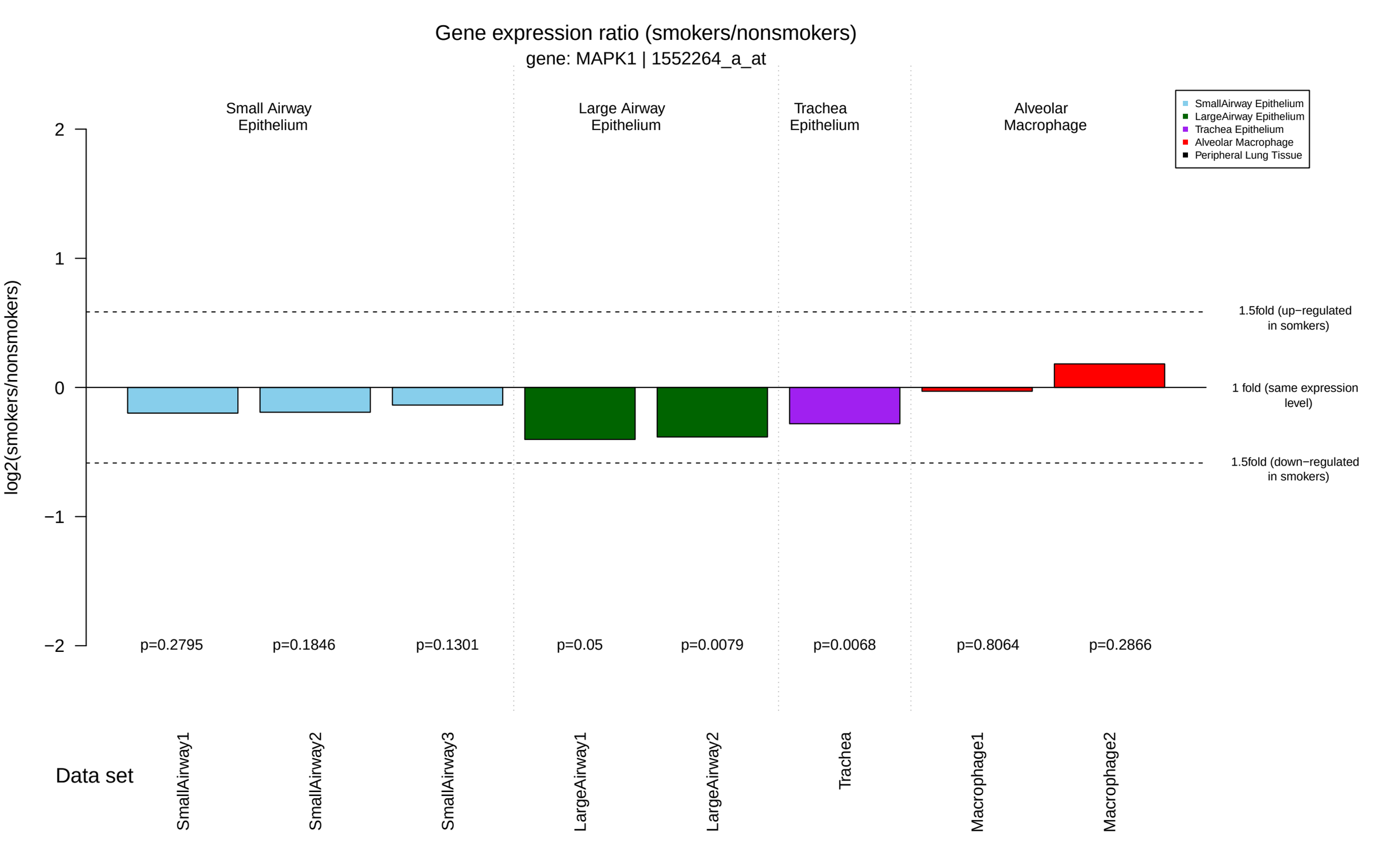
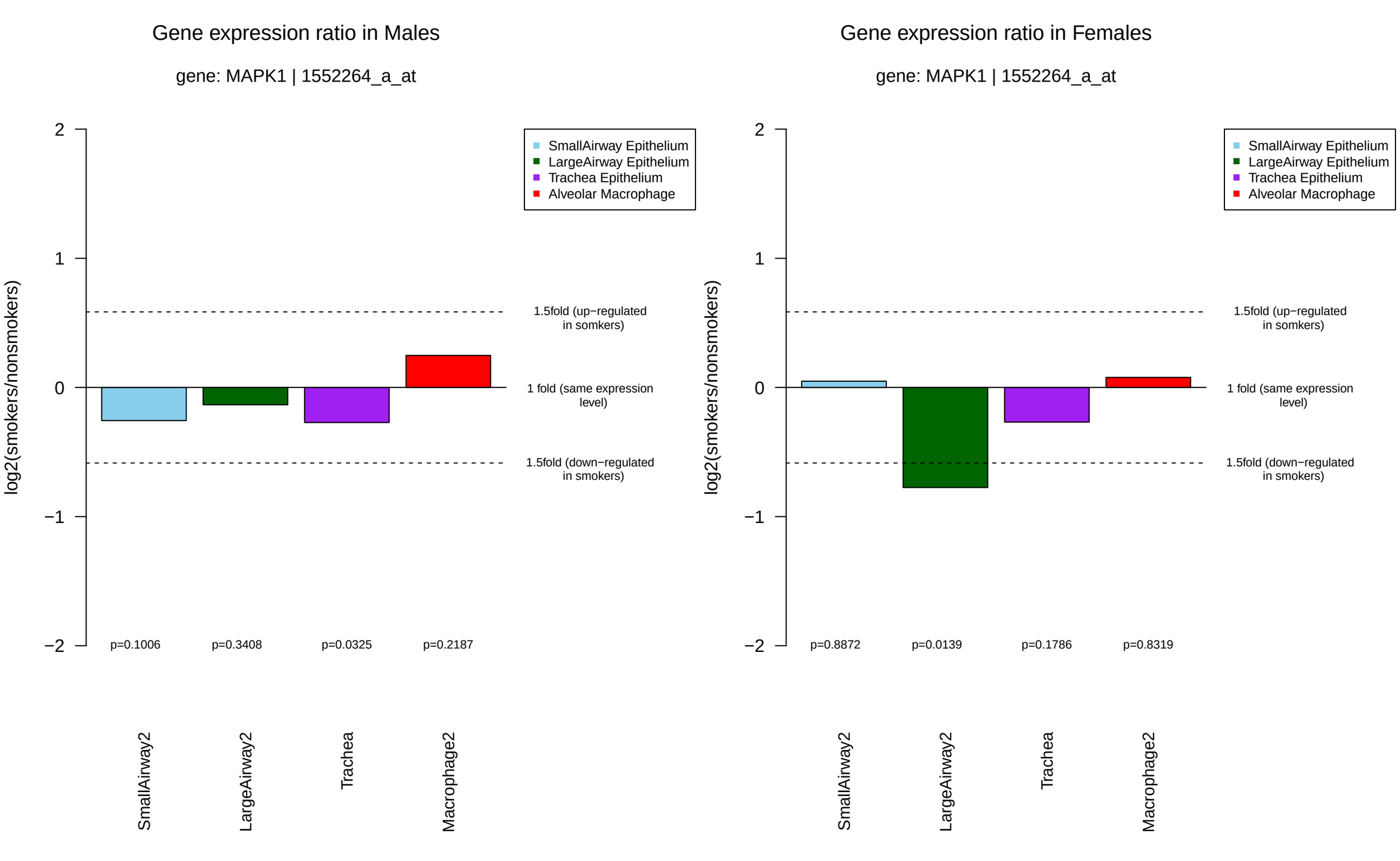
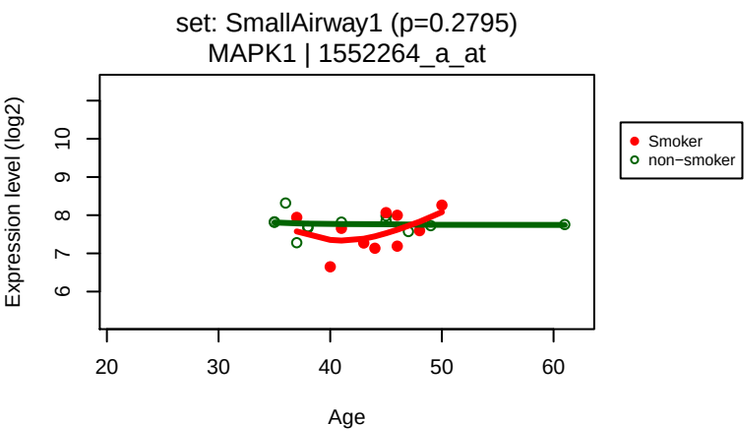
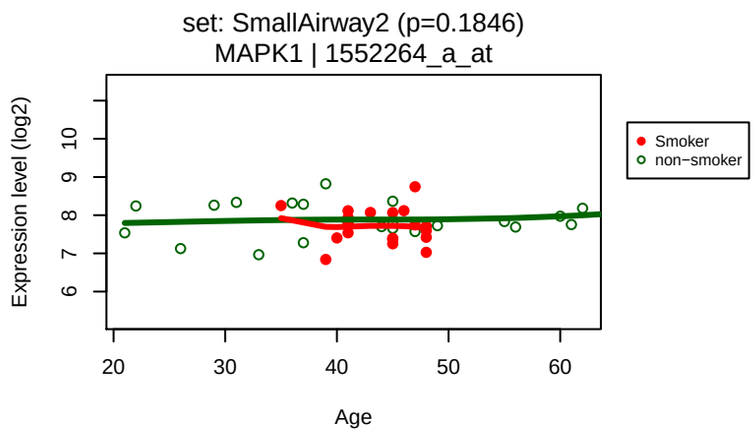
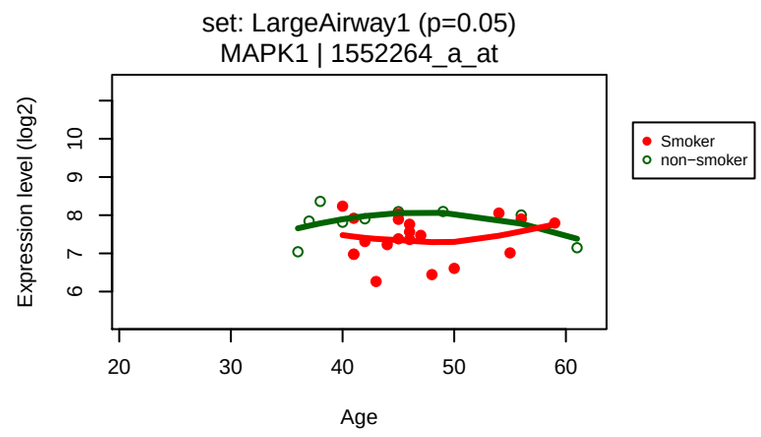

Gene expression of smokers and nonsmokers
gene: MAPK1 | 1552264_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2795
p=0.1846
p=0.1301
p=0.05
p=0.0079
p=0.0068
p=0.8064
p=0.2866
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

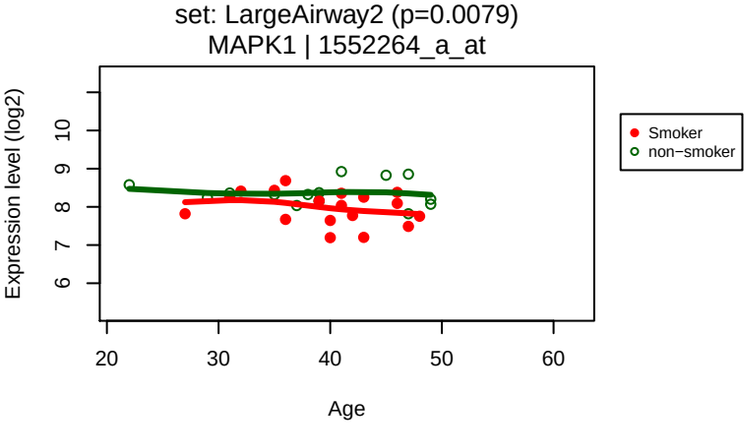

Gene expression ratio (smokers/nonsmokers)
gene: MAPK1 | 1552264_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2795
p=0.1846
p=0.1301
p=0.05
p=0.0079
p=0.0068
p=0.8064
p=0.2866
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

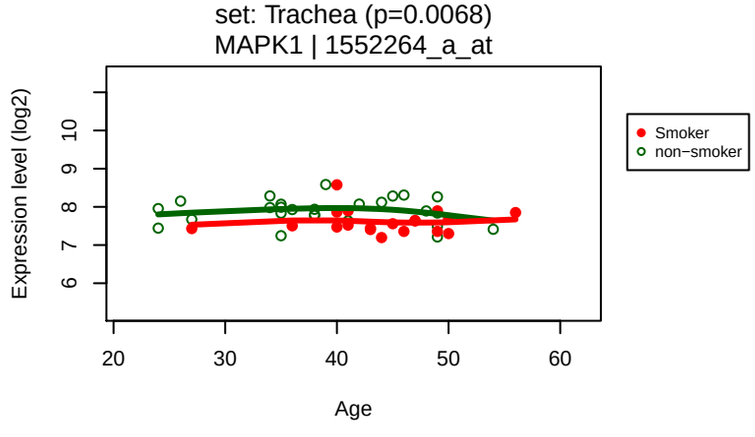

Gene expression ratio in Males
gene: MAPK1 | 1552264_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1006
p=0.3408
p=0.0325
p=0.2187
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAPK1 | 1552264_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8872
p=0.0139
p=0.1786
p=0.8319
SmallAirway2
LargeAirway2
Trachea
Macrophage2

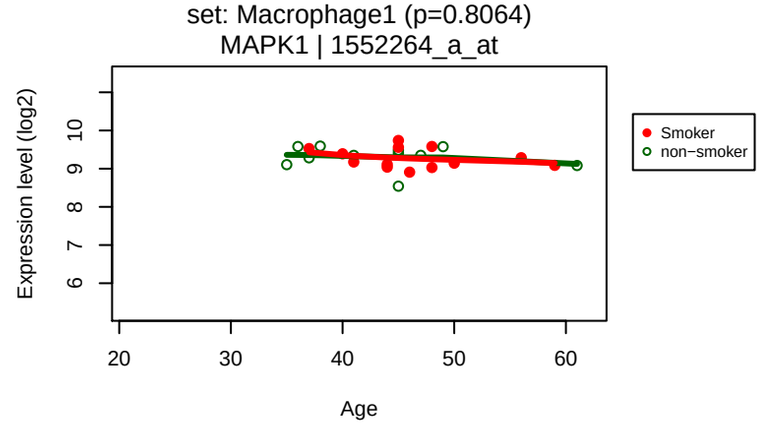

set: SmallAirway1 (p=0.2795)
MAPK1 | 1552264_a_at
Expression level (log2)
Age
Smoker
non−smoker

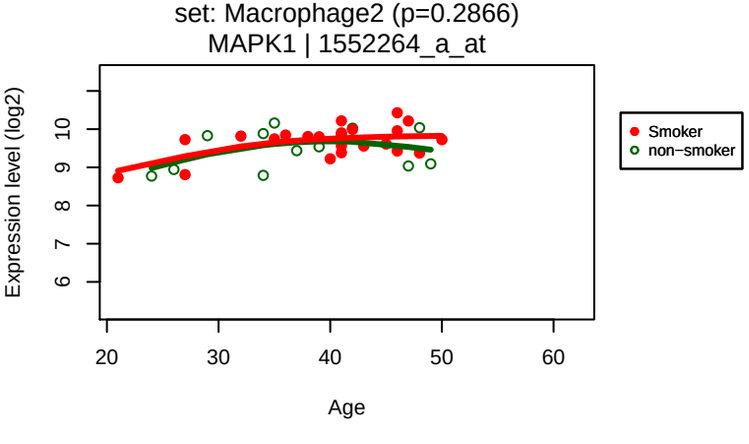

set: SmallAirway2 (p=0.1846)
MAPK1 | 1552264_a_at
Expression level (log2)
Age
Smoker
non−smoker

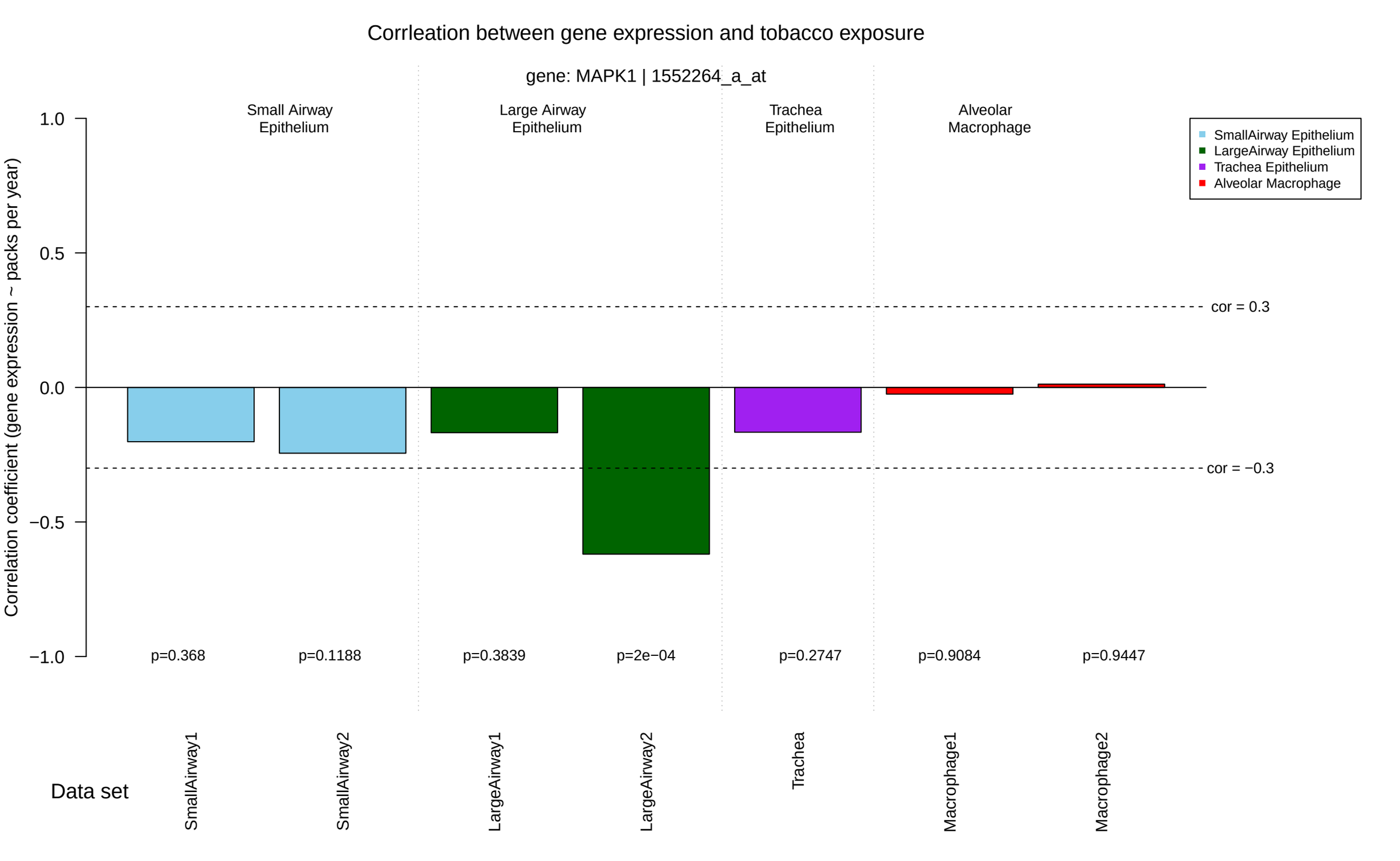

set: LargeAirway1 (p=0.05)
MAPK1 | 1552264_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0079)
MAPK1 | 1552264_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0068)
MAPK1 | 1552264_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8064)
MAPK1 | 1552264_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2866)
MAPK1 | 1552264_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAPK1 | 1552264_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.368
p=0.1188
p=0.3839
p=2e−04
p=0.2747
p=0.9084
p=0.9447
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2